GLOWING EXHAUST MANIFOLD — EMERGENCY

STOP DRIVING · SAFETY CONDITION · DO NOT IDLE IT TO SEE

The manifold is glowing

Bright orange or red exhaust iron means the pipe is hot enough to crack the casting, burn valves, melt a catalyst, destroy oxygen sensors, and ignite oil. That is a stop. Park it outdoors, away from the building and dry grass if anything is smoking. Engine off. Transmission in Park. Parking brake set.

Do this

- Get people away from the wheel well.
- ABC or BC extinguisher within reach. If you see flame, get clear and call the fire department.
- Let it cool completely. An hour is a minimum, not a guarantee. The cast manifold holds heat.
- Do not open the radiator cap. Do not spray water on red iron. Thermal shock cracks it.
- Do not drive it to a shop. Tow it if it has to move.

Glow versus old stain

An active glow is bright orange or red while the engine is running or has just been shut off. Brown, blue, or purple color on a cold manifold is heat stain from some earlier event. Stain is a clue. POSSIBLE past overheat. It is not the same as watching the iron glow. If you saw it glow, that observation stands even after the color fades.

Possible causes — none of these is confirmed

Cause	Why the iron gets hot	Label
Lean combustion	More air than the fuel that was delivered. Combustion temperature and exhaust temperature rise.	POSSIBLE TEST REQUIRED
Misfire, fuel burning in the exhaust	A cylinder that does not fire dumps raw fuel into the manifold. The fuel lights there. This is a classic way to make a manifold glow, and it can happen with a rich cylinder, a dead coil, or a fouled plug.	POSSIBLE TEST REQUIRED
Retarded ignition timing	Late spark pushes burning gas into the exhaust. The 2004 4.0L has a coil rail and no distributor. Timing is not “set” with a wrench. A crank or cam signal problem is the sort of fault that belongs here.	POSSIBLE
Restricted catalyst	A melted mini-cat or main cat holds heat in the manifold. Often the result of a misfire or of oil burning, not the first event. Low power and vacuum that falls as RPM rises are clues.	POSSIBLE TEST REQUIRED

More than one of these can be true. A vacuum leak can start the idle problem. A misfire can be what actually glows the iron. Test both. Do not cut a catalyst out as a test. This is a California-emissions engine. Repair it.

GLOWING MANIFOLD · LEAN PATH AND DAMAGE

Vacuum leak, lean mixture, extreme heat

Working theory, not a closed case. **POSSIBLE**

A vacuum leak adds air. On this MAP engine the idle rises because that air is real. If the injectors and the MAP calculation do not add enough fuel to match it, the mixture goes lean. Lean combustion runs hotter. Exhaust temperature can climb high enough to make the cast manifold glow. That chain is physically possible. It is not established on this truck until fuel trims and upstream oxygen-sensor voltage show a lean condition, and until a misfire and a restricted catalyst have been considered.

What continued running damages

- The manifold itself: cracks at the collector and at the mounting flange.
- Exhaust valves and seats.
- The front and rear mini-catalysts, and the main catalyst. A melted cat then becomes the restriction that keeps the next run just as hot.
- Oxygen sensors. A sensor cooked by a glow, or soaked in the oil that the heat ignited, will not come back.
- Under-hood wiring, the starter, and any oil film. Oil plus a glowing manifold is how this becomes a fire. Guide 06.

After it has cooled — the only tests that are in bounds

- Visual: cracks in the manifold, a cherry-to-gray fresh scale, oil washed onto the iron, a sensor wire melted to the pipe.
- Do not start it for a “short check” until oil has been cleaned off the exhaust and a fire extinguisher is out.
- When a start is allowed: watch from the side, not over the manifold. Idle only. If any orange returns, key off. Tow it.
- Read misfire counters if the scan tool has them. Read upstream O2 and fuel trims. A single glowing event with a dead cylinder is a misfire case until proven otherwise.
- Catalyst check, only on a later and cooler run: infrared temperature ahead of and behind each cat, or a pressure gauge in front of the cat. Do this after the idle and the oil leak are under control, not on the day the manifold was red.

Return-to-service rule. The manifold stayed dark on a full warm idle, oil is not reaching the pipe, warm idle is in the 700–800 rpm screen, and a rescan shows no new misfire or fuel-control fault. If any one of those is missing, it is not finished. If the glow returns, stop immediately. Do not reset anything and try again.